

Module E: Fun Rock-Paper-Scissors

Development Challenge

(Non-scoring module, 90-minute time limit for development)

Exam Instructions

- **Theme** : Fun Rock-Paper-Scissors
- **Duration** : 90 minutes
- **Rules** : Throughout the development process, developers are free to access online resources and utilize AI tools for assistance.
- **Objective** : To complete the basic game logic and creative visual effects, ultimately enabling the work to be displayed to an audience on a large screen.
- **Scoring Notes** : This module does not contribute to the overall competition score; its core objectives are creativity and fun.

Task Requirements

I. Basic Required Items (Must be completed)

1. Core game logic

- Implement player vs. computer battle mode
- Supports three gesture options: rock, paper, scissors.
- The winning logic is: Rock beats Scissors, Scissors beats Paper, Paper beats Rock.
- Draw and Win/Loss Result Determination and Tips
- Game reset/play another game function

2. Basic Interaction and Visuals

- Clear gesture selection buttons (text/icon/image are all acceptable)
- The display effect of computer randomly throwing punches
- Results feedback (text prompts + win/loss highlighted)
- Adapted for large screen displays, with no obvious interface misalignment.

II. Creative Bonus Points (Experimentation is encouraged; the more fun, the better)

(You can choose any 2-3 items below to implement, or feel free to use other creative ideas.)

- **Theme skins** : Customizable gesture icons and background styles (such as cartoon IPs, event elements, and dynamic backgrounds).
- **Interactive animations** : Click feedback when punching, special effects for winning or losing (zooming in, fade-in, particle effects).
- **Fun statistics** : Records data from multiple matches (player wins, computer wins, number of draws, winning streaks).
- **Sound feedback** : Add simple sound effects (such as selection sound effect, victory sound effect) for punches, wins and losses, and resets.
- **Fun Easter Eggs** : Special effects with hidden trigger conditions (such as an encouraging animation appearing after losing 3 games in a row, and a special effect triggered after winning 3 games in a row).
- **Fun AI** : Add "personas" to computer AI (such as a bias towards a certain gesture or changes to the starting prompts).

Development and Submission Requirements

1. **Programming language/framework** : Not limited
2. **Material Acquisition** : You are free to use publicly available online materials and AI-generated images/sound effects without commercial licensing.
3. **Submission method** : After completion, submit the project files (including source code and operating instructions) to the designated Module E Gitea repository.
4. **Presentation Preparation** : Reserve 1-2 minutes to introduce your creative ideas and implementation process to the audience/judges.

Friendly Reminder

- With 90 minutes allotted, it's recommended to quickly complete the basic battle logic first, and then gradually add creative effects.
- Problems can be solved at any time using online and AI tools; the key is to create works that are "fun and have your own unique characteristics."

- When showcasing your work, highlight 1-2 of your most brilliant creative ideas so that the audience can quickly grasp your ingenuity .